

Kidus Assefa

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SUMMARY

AI-native builder and incoming M.S. in AI Systems Management candidate at Carnegie Mellon University (expected graduation December 2027), targeting the AI Native Summer Analyst role. Hands-on experience designing, testing, and iterating on LLM-powered systems, RAG pipelines, agents, fine-tuned models, and safety/guardrail layers built with the Anthropic Claude API. Blends applied AI engineering (prompt engineering, system instructions, iterative refinement, AI integration) with a cyber-defense foundation (GCIH, GSEC) and a track record of turning analysis into clear, data-driven recommendations for stakeholders.

EDUCATION

Carnegie Mellon University

M.S., AI Systems Management

Aug 2026 - Dec 2027

Pittsburgh, PA

- **Achievements:** 50% Merit Scholarship

SANS Technology Institute

Undergraduate Certificate, Applied Cybersecurity (ACS)

Oct 2025 - Aug 2026

Online

- **Coursework:** SEC301 (GFACT), SEC401 (GSEC), SEC504 (GCIH)

Virginia Tech

B.S., Cybersecurity Management & Analytics

Aug 2022 - Dec 2024

Blacksburg, VA

CERTIFICATIONS

- CodePath Applications of AI Engineering (AI 201)
- GIAC Certified Incident Handler (GCIH)
- CodePath Foundations of AI Engineering (AI 101)
- GIAC Security Essentials (GSEC)
- GIAC Foundational (GFACT)
- CompTIA Security+

Nov 2026

May 2026

May 2026

Feb 2026

Aug 2025

May 2024

WORK EXPERIENCE

CodePath

Tech Fellow

Jan 2026 - Present

- Mentor and troubleshoot for 100+ learners across hands-on cyber-defense labs, collaborating with instructors to unblock students in real time.
- Administer Azure VM and Slack infrastructure supporting 100+ learners across concurrent lab environments.
- Deliver labs on reconnaissance, credential harvesting, and social-engineering attack chains; drove a 100% capstone completion rate.

New River Community College

Education Support Specialist

Jul 2021 - May 2026

- Guided students through networking, programming, and SQL coursework, translating complex IT concepts into clear explanations for non-technical learners.
- Facilitated Math and IT tutoring, improving student performance by 15% across sessions.

Virginia Tech & Commonwealth Cyber Initiative

Summer Research Intern

May 2024 - Aug 2024

- Applied Python-based threat analysis to public datasets and identified 3 compliance gaps that directly shaped a federal cybersecurity certification framework.
- Translated findings into data-driven recommendations and presented privacy-risk mitigations to stakeholders, informing regulatory compliance strategy.

PROJECTS

PawPal+ AI

Feb 2026 - Present

- AI Output Validation & Guardrail Architecture
- Built an LLM-powered application integrating the Anthropic Claude API with rule-based scheduling to automate decisions from structured user data.
- Ran prompt-engineering and system-instruction experiments, iteratively refining outputs to improve response accuracy and reliability.
- Designed a responsible-AI guardrail layer that validates model responses and triggers safe fallbacks - principles transferable to AI threat-analysis and validation pipelines.

Applications of AI Engineering (AI 201)

2026

CodePath

- Built and tested a Retrieval-Augmented Generation (RAG) system to ground LLM responses in external knowledge and improve accuracy.
- Designed and tested an autonomous agent, implementing agentic workflows with tool use and multi-step reasoning.
- Fine-tuned and evaluated a model, benchmarking performance gains against a baseline.
- Implemented an AI safety layer enforcing responsible-AI guardrails on model outputs.
- 10-week program, 2026

Incident Response Lab Practice

- GCIH Preparation
- Completed 50+ hours of hands-on IR labs: host and network investigation, malware analysis, log analysis, and threat containment across Windows/enterprise environments.

SKILLS

- **AI & LLM Engineering:** Prompt Engineering, System Instructions, LLMs, RAG, Agentic Workflows, Fine-Tuning, AI Integration, Anthropic Claude API, Responsible AI / AI Validation, Python, SQL, API Integration
- **Cyber Defense & IR:** Incident Response, Threat Analysis, Network Defense, Security Operations, Ethical Hacking, OSINT
- **Tools & Infrastructure:** Azure VM Administration, Linux, Web Development, Microsoft Office
- **Core Competencies:** Problem Solving, Collaboration, Communication, Adaptability, Learning Agility